dy/dx

y' + Py = Q

12M21 Differential Equation

Quick Revision Handout

∫ dx

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Differential Equation: Basics

Definition

An equation that contains the independent variable, the dependent variable and **derivatives** of the dependent variable.

$$\frac{dy}{dx} + y = \sin x$$

Types

- **Ordinary (ODE):** only one independent variable.
- **Partial (PDE):** more than one independent variable.

JEE deals with ODEs only.

Order

Order of the **highest-order derivative** present in the equation.

Degree

Power of the highest-order derivative **after** the equation is made a polynomial in its derivatives (no radicals or fractions of derivatives).

Degree not defined when the derivatives sit inside functions that cannot be removed, e.g. $\sin y'$, $\cos y''$.

Order & Degree: Examples

Equation	Order	Degree
$\frac{dy}{dx} + y = x$	1	1
$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 = x$	2	3
$y = x\frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$	1	2
$k\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}$	2	2
$\left(\frac{d^3y}{dx^3}\right)^{2/3} = \frac{d^2y}{dx^2}$	3	2
$\frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$	2	not defined

Steps

1. Remove radicals and fractional powers of derivatives (isolate and raise to a power).
2. Highest derivative gives the **order**.
3. Its power gives the **degree**.

Remember: order and degree are always positive integers.

Formation of a Differential Equation

n independent arbitrary constants $\Longrightarrow$ DE of order **n**

Method

1. Differentiate the family **n** times.
2. Eliminate the **n** constants from the **n+1** equations.

Family	Differential equation
$y = mx$	$y = x\dfrac{dy}{dx}$
$x^2 + y^2 = a^2$	$x + y\dfrac{dy}{dx} = 0$
$y^2 = 4ax$	$y = 2x\dfrac{dy}{dx}$
$y = Ae^{2x} + Be^{-3x}$	$y'' + y' - 6y = 0$

Watch out: constants must be **independent**. In $y = ae^{x+b}$ we have $y = ce^x$ (with $c = ae^b$), so there is only **one** constant and the order is 1.

Solution of a Differential Equation

General solution

Contains **n independent arbitrary constants** (n = order). It represents a **family of curves**.

Particular solution

Constants given definite values using conditions. It represents **one curve** from the family.

Singular solution

A solution that **cannot** be obtained from the general solution by any choice of constants.

Example: $\dfrac{dy}{dx} = 2\sqrt{y} \Longrightarrow$ general: $y = (x + C)^2$.
$y = 0$ also satisfies it, but no C gives it: **singular**.

Initial value problem

$\dfrac{dy}{dx} = 3x^2,\ y(1) = 2 \Longrightarrow y = x^3 + C$, and $2 = 1 + C$ gives $C = 1$, so $y = x^3 + 1$.

Tip: use the given condition at the **end**, after integrating, to find C.

Variable Separation Method

When the equation can be written as

$$\frac{dy}{dx} = f(x)\,g(y)$$

separate the variables, then integrate:

$$\int \frac{dy}{g(y)} = \int f(x)\,dx + C$$

Examples

1. $\dfrac{dy}{dx} = e^{x-y} \implies e^y dy = e^x dx \implies \boldsymbol{e^y = e^x + C}$

2. $\dfrac{dy}{dx} = \dfrac{1+y^2}{1+x^2} \implies \boldsymbol{\tan^{-1} y - \tan^{-1} x = C}$

3. $\dfrac{dy}{dx} = xy,\ y(0) = 1 \implies \ln y = \dfrac{x^2}{2} + C$, and $C = 0$, so $\boldsymbol{y = e^{x^2/2}}$

Tips:

- Dividing by $g(y)$ can lose constant solutions where $g(y) = 0$. Check them.
- Write the constant as $\ln C$ when logs appear.

Homogeneous Differential Equation

$$\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)} = F\left(\frac{y}{x}\right)$$

f, g homogeneous of the same degree: $f(\lambda x, \lambda y) = \lambda^n f(x,y)$.

Method

1. Put $y = vx$, so $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.
2. Separate the variables:

$$\frac{dv}{F(v) - v} = \frac{dx}{x}$$

3. Integrate, then put back $v = \dfrac{y}{x}$.

If the form is $\dfrac{dx}{dy} = F\left(\dfrac{x}{y}\right)$, put $x = vy$ instead.

Ex. $\dfrac{dy}{dx} = \dfrac{x^2+y^2}{2xy}$. With $y = vx$: $x\dfrac{dv}{dx} = \dfrac{1-v^2}{2v}$, so $\dfrac{2v\,dv}{1-v^2} = \dfrac{dx}{x}$, giving $\boldsymbol{x^2 - y^2 = Cx}$

Reducible to Homogeneous

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$$

Case 1: $\dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2}$

Lines intersect at (h, k). Put $x = X + h,\ y = Y + k$. The equation becomes **homogeneous in** X, Y.

Ex. $\dfrac{x - y + 1}{x + y - 3}$: the lines meet at $(1, 2)$, so use $x = X + 1,\ y = Y + 2$ to get $\dfrac{dY}{dX} = \dfrac{X - Y}{X + Y}$.

Case 2: $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2}$

Lines are parallel. Put $z = a_1x + b_1y$. The equation becomes **variable separable** in x, z.

Ex. $\dfrac{dy}{dx} = \dfrac{x + y + 1}{x + y - 1}$. Put $z = x + y$: $\dfrac{dz}{dx} = \dfrac{2z}{z - 1}$, so $\dfrac{z - 1}{2z}dz = dx$, giving $\boldsymbol{y - x - \ln|x + y| = C}$

Linear Differential Equation

$$\frac{dy}{dx} + P\,y = Q \qquad (P, Q \text{ are functions of } x)$$

$$\text{I.F.} = e^{\int P\,dx}$$

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.})\,dx + C$$

Linear in x

$$\frac{dx}{dy} + P\,x = Q \quad (P, Q \text{ functions of } y)$$

$$\text{I.F.} = e^{\int P\,dy}, \qquad x \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.})\,dy + C$$

Ex.1 $\dfrac{dy}{dx} + \dfrac{y}{x} = x^2$. I.F. $= x$, so $xy = \dfrac{x^4}{4} + C$, i.e. $\boldsymbol{y = \dfrac{x^3}{4} + \dfrac{C}{x}}$

Ex.2 $(x + y)\dfrac{dy}{dx} = 1$ gives $\dfrac{dx}{dy} - x = y$. I.F. $= e^{-y}$, so $\boldsymbol{x = Ce^y - y - 1}$

Tip: do **not** add a constant while finding the I.F.

Reducible to Linear + I.F. Table

Bernoulli's equation

$$\frac{dy}{dx} + Py = Q\,y^n \qquad (n \neq 0, 1)$$

Divide by y^n and put $z = y^{1-n}$. It becomes linear:

$$\frac{dz}{dx} + (1-n)P\,z = (1-n)Q$$

Ex. $\frac{dy}{dx} + \frac{y}{x} = y^2$. Put $z = \frac{1}{y}$: $\frac{dz}{dx} - \frac{z}{x} = -1$, giving $\boldsymbol{\frac{1}{xy} = C - \ln|x|}$

Also: if $f'(y)\frac{dy}{dx} + P\,f(y) = Q$, put $f(y) = z$.

Quick Integrating Factors

$P(x)$	I.F.	$P(x)$	I.F.
a	e^{ax}	$\tan x$	$\sec x$
$\frac{1}{x}$	x	$\cot x$	$\sin x$
$-\frac{1}{x}$	$\frac{1}{x}$	$\frac{1}{1+x^2}$	$e^{\tan^{-1}x}$
$\frac{k}{x}$	x^k	$\frac{f'(x)}{f(x)}$	$f(x)$

General Form of Variable Separation

A. Substitution

$\frac{dy}{dx} = f(ax+by+c)$: put $z = ax+by+c$, then $z' = a + b\,f(z)$.

$y\,f(xy)\,dx + x\,g(xy)\,dy = 0$: put $xy = v$.

B. Exact differentials to spot

$x\,dy + y\,dx$	$= d(xy)$
$\frac{x\,dy - y\,dx}{x^2}$	$= d\left(\frac{y}{x}\right)$
$\frac{y\,dx - x\,dy}{y^2}$	$= d\left(\frac{x}{y}\right)$
$\frac{x\,dy - y\,dx}{xy}$	$= d\left(\ln\frac{y}{x}\right)$
$\frac{x\,dy - y\,dx}{x^2+y^2}$	$= d\left(\tan^{-1}\frac{y}{x}\right)$
$x\,dx + y\,dy$	$= \frac{1}{2}\,d(x^2+y^2)$

Ex.1 $\frac{dy}{dx} = (x+y)^2$: put $z = x+y$, so $\frac{dz}{dx} = 1+z^2$ and $\boldsymbol{x+y = \tan(x+C)}$

Ex.2 $x\,dy - y\,dx = (x^2+y^2)dx$ gives $d\left(\tan^{-1}\frac{y}{x}\right) = dx$, so $\boldsymbol{\tan^{-1}\frac{y}{x} = x + C}$

Applications: Geometry

Let $y' = \dfrac{dy}{dx}$ at the point $P(x, y)$.

Slope of tangent $= y'$ | Slope of normal $= -\dfrac{1}{y'}$

Tangent: $Y - y = y'(X - x)$

Length of tangent	$\left\lvert \dfrac{y\sqrt{1+y'^2}}{y'} \right\rvert$
Length of normal	$\left\lvert y\sqrt{1+y'^2} \right\rvert$
Subtangent	$\left\lvert \dfrac{y}{y'} \right\rvert$
Subnormal	$\lvert y\,y' \rvert$

Orthogonal trajectories

1. Form the DE of the given family (eliminate c).
2. Replace $\dfrac{dy}{dx}$ by $-\dfrac{dx}{dy}$.
3. Solve the new DE.

Ex. $y = cx$ gives $y' = \dfrac{y}{x}$. Replacing: $-\dfrac{dx}{dy} = \dfrac{y}{x}$, so $x\,dx + y\,dy = 0$, giving $\boldsymbol{x^2 + y^2 = C}$

Applications: Rate Problems

Growth & decay

$$\frac{dN}{dt} = kN \;\Rightarrow\; N = N_0 e^{kt}$$

$k > 0$ growth, $k < 0$ decay. Half-life $T_{1/2} = \dfrac{\ln 2}{|k|}$.

Newton's law of cooling

$$\frac{dT}{dt} = -k\,(T - T_s) \;\Rightarrow\; T = T_s + (T_0 - T_s)e^{-kt}$$

T_s = surroundings, T_0 = initial temperature.

Mixing (tank) problems

$\dfrac{dQ}{dt}$ = (rate in) − (rate out), where rate out $= \dfrac{Q}{V} \times$ outflow rate.

Ex.1 Bacteria double in 3 h. Then $2 = e^{3k}$, so 8-fold needs e^{9k}: **9 h**.

Ex.2 A body at 100° in a 20° room cools to 60° in 10 min. Then $40 = 80e^{-10k}$, so $e^{-10k} = \frac{1}{2}$. At 20 min: $T = 20 + 80 \cdot \frac{1}{4} =$ **40°**

Steps: write the statement as a DE in t, solve, then use both conditions to find the constant and k.

Method Chooser Chart

Type	Identify	Do this
Separable	$f(x)g(y)$	separate and integrate
$f(ax+by+c)$	$y' = f(ax+by+c)$	$z = ax+by+c$
Homogeneous	$y' = F(y/x)$	$y = vx$
Lines form	$\dfrac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$	shift origin, or $z = a_1x+b_1y$
Linear in y	$y' + Py = Q$	I.F. $= e^{\int P\,dx}$
Linear in x	$x' + Px = Q(y)$	I.F. $= e^{\int P\,dy}$
Bernoulli	$y' + Py = Qy^n$	$z = y^{1-n}$
Differential forms	$x\,dy \pm y\,dx$ patterns	use $d(\cdot)$ results

Tips & Common Mistakes

Avoid these

- Forgetting $+C$ after integrating.
- Dropping the modulus in $\ln|x|$.
- Losing constant solutions while dividing.
- Adding a constant inside the I.F.
- Wrong sign of P after writing the standard form.
- Not clearing radicals before finding degree.

Standard integrals used here

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$$

$$e^{\ln f(x)} = f(x)$$

Before submitting

- Match the type first, then pick the method.
- Apply initial conditions only at the end.
- Differentiate your answer to verify it.